

Yissum Research Intelligence Report — August 2026

8 high-potential paper(s) · Monthly · August 2026 · Generated August 01, 2026

Hebrew University's latest research highlights significant commercial potential across diverse sectors. Breakthroughs in biotechnology, including smart gene therapies for neurological disorders and RNAi-based crop protection, offer solutions for pressing global health and food security challenges. Further advancements in materials science present opportunities in high-value defense and electronics markets, while AI-driven drug discovery platforms promise personalized medicine innovations. These developments provide compelling near-term investment opportunities for industry partners seeking disruptive technologies.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Roy Maman** — Materials, Imaging
- **Osnat Malka** — AgriTech, Genomics, Synthetic Biology
- **Tawfeeq Shekh-Ahmad** — Drug Discovery, Neuroscience, Genomics, Clinical
- **Rachel Nechushtai** — Drug Discovery, Proteomics, Clinical
- **Tommy Kaplan** — Genomics, Drug Discovery, Diagnostics, Software/AI
- **Dina Schneidman-Duhovny** — Drug Discovery, Immunology, Software/AI, Synthetic Biology
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics
- **Esti Davidovich** — Medical Device, Clinical

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Metasurface platform for simultaneous and uncorrelated emissivity control over MWIR and LWIR spectral bands.

32/50

"Novel metasurface independently manipulates thermal signatures for stealth, advanced thermal management, and specialized sensors."

Main researcher: Roy Maman · Roy.maman@mail.huji.ac.il.

Institute of Applied Physics, The Faculty of Science, The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Jerusalem, 91904, Israel.

Published in Light, science & applications · 2026 Jul 23

ABOUT THIS RESEARCH

Researchers have developed a metasurface capable of independently controlling how an object emits heat (thermal radiation) across two different infrared spectrums. This allows for the precise manipulation of thermal signatures and heat dissipation without one wavelength affecting the other.

COMMERCIAL ANGLE

High-value applications in stealth technology, advanced thermal management for electronics, and specialized non-invasive thermal imaging sensors.

COMMERCIALISATION METRICS

Novelty

7/10

The ability to achieve uncorrelated, simultaneous control over two distinct infrared bands (MWIR and LWIR) represents a significant functional leap over single-band or correlated metasurfaces, though it builds on existing plasmonic/dielectric resonator principles.

Commercial Potential

7/10

The ability to independently control MWIR and LWIR emissivity creates a high-value platform for thermal camouflage, energy-efficient coatings, and advanced sensing. While the technical application is clear, the score is tempered by the lack of provided data on scalability and manufacturing costs.

Market Size

7/10

The ability to independently control MWIR and LWIR emissivity is a platform-enabling capability with high value in defense (stealth/cloaking) and industrial thermal management, though it targets specialized high-end niches rather than a mass consumer market.

Tech Readiness

3/10

The paper describes the development of a metasurface platform for spectral control, which is characteristic of fundamental materials science and lab-stage proof-of-concept work. There is no evidence of integration into a commercial device, field testing, or scalable manufacturing processes.

IP Strength

8/10

The development of a platform for uncorrelated spectral control in two distinct bands represents a high-value 'platform-enabling' innovation with strong potential for broad patent claims across defense and industrial sensing.

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2. Creating resistance to the whitefly *Bemisia tabaci* in cassava through RNAi-mediated targeting of multiple insect metabolic processes. 31/50

"RNAi-engineered cassava offers robust whitefly resistance, securing vital food crops and opening biopesticide avenues."

Main researcher: Osnat Malka

Published · 2026-05

ABOUT THIS RESEARCH

Researchers developed a way to make cassava plants resistant to whiteflies, which are the primary carriers of devastating diseases in East Africa. They used RNA interference (RNAi) to target and disrupt essential metabolic processes within the insects.

COMMERCIAL ANGLE

Development of genetically engineered, pest-resistant crop seeds and RNAi-based biopesticides to reduce yield loss in global cassava production.

COMMERCIALISATION METRICS

Novelty

6/10

While targeting multiple metabolic pathways via RNAi is a robust strategy, host-induced gene silencing (HIGS) for pest control is a well-established technique; the novelty here lies primarily in the application to a specific crop-pest system rather than a fundamental scientific breakthrough.

Commercial Potential

7/10

The technology addresses a massive, billion-dollar agricultural pain point with a clear product path (transgenic resistant cultivars). However, the commercial potential is slightly tempered by the regulatory hurdles of GMOs in African markets and the necessity for multi-gene targeting to prevent insect resistance.

Market Size

8/10

The target market is vast and economically critical, with annual losses exceeding \$1.25 billion and direct impact on food security for millions in East Africa. The RNAi approach offers a scalable, platform-like solution to a 'super-abundant' pest affecting a primary global staple crop.

Tech Readiness

3/10

The research is in the early laboratory stage, focusing on the identification of candidate genes and the production of RNAi constructs, which precedes field testing and regulatory approval.

IP Strength

7/10

The use of multi-gene RNAi targeting specific metabolic pathways allows for strong composition-of-matter patents on the constructs and potential 'stacking' strategies to prevent resistance. However, defensibility is slightly tempered by the general nature of RNAi techniques and potential regulatory hurdles in agricultural biotech.

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3. Activity-dependent antioxidant gene therapy protects against epileptogenesis. 31/50

"Smart gene therapy prevents epilepsy by activating antioxidant enzymes only during brain activity, minimizing side effects."

Main researcher: Tawfeeq Shekh-Ahmad · Tawfeeq.Shekh-Ahmad@mail.huji.ac.il.

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Tawfeeq.Shekh-Ahmad@mail.huji.ac.il.

Published in Signal transduction and targeted therapy · 2026-06

ABOUT THIS RESEARCH

This research describes a gene therapy approach that uses natural brain activity to trigger the production of antioxidant enzymes. This mechanism aims to prevent the development of epilepsy following an initial brain injury.

COMMERCIAL ANGLE

Development of smart, responsive gene therapies for neurological disorders that activate only during specific physiological states to minimize side effects.

COMMERCIALISATION METRICS

Novelty

7/10

The work is highly innovative for shifting the therapeutic paradigm from reactive seizure control to proactive, activity-dependent gene modulation to prevent epileptogenesis. However, while the mechanism is sophisticated, antioxidant gene therapy is an established concept, making the novelty incremental rather than a complete disruption of biological principles.

Commercial Potential

7/10

The technology targets a high-unmet-need clinical indication (epileptogenesis) with a specific mechanism (antioxidant gene therapy), offering a clear path toward a specialized therapeutic product. However, the lack of abstract details regarding delivery vectors and scalability limits a higher score.

Market Size

7/10

The target market for epilepsy therapeutics is massive and high-value, representing a multi-billion dollar global neurology segment. However, the score is tempered by the specific niche of epileptogenesis prevention, which is a subset of the broader epilepsy market and faces complex clinical adoption hurdles.

Tech Readiness

3/10

The title suggests a fundamental mechanistic study using gene therapy in a disease model, which typically implies preclinical, lab-based proof-of-concept research rather than a developed therapeutic candidate.

IP Strength

7/10

The use of activity-dependent promoters for targeted gene therapy provides a strong basis for composition-of-matter and method-of-use patents, though defensibility depends on the novelty of the specific promoter/transgene combination.

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4. Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein. 31/50

"Breakthrough peptide targets and selectively destroys primary and metastatic ovarian cancer cells with low toxicity."

Main researcher: Rachel Nechushtai · rachel@mail.huji.ac.il.

Institute of Life Science, Faculty of Science and Mathematics, The Hebrew University of Jerusalem, Edmond J. Safra Campus at Givat Ram, Jerusalem 9190401, Israel. Electronic address: rachel@mail.huji.ac.il.

Published in Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie · 2026 Jul 1

ABOUT THIS RESEARCH

Researchers have developed a specialized peptide derived from the NAF-1/CISD2 protein that selectively destroys primary and metastatic ovarian cancer cells. The treatment showed efficacy and low toxicity in both laboratory settings and human ovarian cancer mouse models.

COMMERCIAL ANGLE

Development of a first-in-class targeted peptide therapeutic to address the high relapse rates and low survival outcomes of advanced ovarian cancer.

COMMERCIALISATION METRICS

Novelty 6/10

The use of a protein-derived peptide for selective cancer cytotoxicity is a known strategy; however, targeting the NAF-1/CISD2 pathway specifically for ovarian cancer represents a novel application of this protein's biology.

Commercial Potential 7/10

The peptide addresses a high-unmet-need market (refractory ovarian cancer) with demonstrated selectivity and in vivo efficacy, offering a clear path to a therapeutic product. However, the score is tempered by the typical pharmacological challenges of peptides, such as stability and delivery, which are not addressed in the abstract.

Market Size 8/10

Ovarian cancer represents a high-value target market due to a massive global incidence rate and a critical lack of effective treatments for relapsed or metastatic cases. The addressable demand is significant given the high relapse rate (80%) and poor 5-year survival outcomes.

Tech Readiness 3/10

The technology is at a basic proof-of-concept stage, having demonstrated efficacy in vitro and in xenograft mouse models, which corresponds to early preclinical development.

IP Strength 7/10

The innovation centers on a specific peptide sequence derived from a human protein (NAF-1/CISD2), which is typically highly patentable as a composition of matter. While the specificity for ovarian cancer provides strong defensibility, the score is tempered by the inherent vulnerability of peptides to proteolysis and potential ease of sequence modification by competitors.

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5. The genetic basis for DNA methylation variation across tissues and development.

30/50

"High-resolution epigenetic atlas maps how genetic variations control DNA methylation, enabling precise biomarkers and drug targets."

Main researcher: Tommy Kaplan · tommy@cs.huji.ac.il.

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel. tommy@cs.huji.ac.il; Dept. of Developmental Biology and Cancer Research, Institute for Medical Research Israel-Canada, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. tommy@cs.huji.ac.il; Center for Computational Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. tommy@cs.huji.ac.il; Centre for Cancer Cell and Molecular Biology, Barts Cancer Institute, Queen Mary University of London, London, UK. tommy@cs.huji.ac.il; Centre for Epigenetics, Queen Mary University of London, London, UK. tommy@cs.huji.ac.il.

Published in Nature communications · 2026-04

ABOUT THIS RESEARCH

Researchers have mapped how specific genetic variations control DNA methylation patterns across different cell types and developmental stages in both mice and humans. This study identifies thousands of locations where DNA sequence changes drive epigenetic programming, linking genetics directly to gene regulation and disease risk.

COMMERCIAL ANGLE

This high-resolution atlas of sequence-dependent methylation can be leveraged to develop highly precise epigenetic biomarkers for early disease detection and to identify new regulatory targets for drug discovery.

COMMERCIALISATION METRICS

Novelty

9/10

The paper introduces a groundbreaking unifying framework that connects genetic sequence to epigenetic programming across species and developmental stages. It provides a massive, high-resolution atlas that transitions the field from descriptive epigenetics to a mechanistic understanding of how SNPs drive tissue-specific methylation.

Commercial Potential

6/10

While the atlas provides high-value foundational data for drug target discovery and disease mechanism modeling, it is a descriptive resource rather than a direct therapeutic or diagnostic product. Its commercial strength lies in its potential as a platform-enabling tool for biotech R&D rather than a standalone commercial offering.

Market Size

8/10

The atlas serves as a high-value, platform-enabling resource for drug discovery and functional genomics, potentially addressing the massive multi-billion dollar markets for precision medicine and target identification. By mapping the link between SNPs and methylation, it provides actionable data for interpreting disease-associated variants, a critical bottleneck in modern pharmaceutical R&D.

Tech Readiness

3/10

This is fundamental discovery science focused on mapping biological mechanisms and creating a descriptive atlas rather than developing a specific tool, diagnostic, or therapeutic intervention. While the data is high-value for downstream drug discovery, the research itself remains at the basic mechanistic understanding stage.

IP Strength

4/10

The paper is primarily fundamental discovery science that maps natural biological correlations rather than describing a proprietary tool, method, or molecule. While the resulting atlas is highly valuable for drug discovery, the underlying genetic and epigenetic associations themselves are natural phenomena and generally considered non-patentable.

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6. Progress in structure prediction and design of adaptive immune receptors.

30/50

"AI-driven platform advances precision design of synthetic immune receptors for personalized cancer immunotherapies."

Main researcher: Dina Schneidman-Duhovny · dina.schneidman@mail.huji.ac.il.

The Rachel and Selim Benin School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel.

Electronic address: dina.schneidman@mail.huji.ac.il.

Published in Current opinion in structural biology · 2026 Jul 10

ABOUT THIS RESEARCH

The paper reviews advancements in using computational tools to predict the shapes and design the functions of immune receptors. This allows scientists to better understand how the body recognizes pathogens and how to engineer custom receptors for therapy.

COMMERCIAL ANGLE

Developing AI-driven platforms for the precision design of synthetic antibodies and T-cell receptors for personalized cancer immunotherapies.

COMMERCIALISATION METRICS

Novelty

4/10

Based on the title, this is a review paper summarizing 'progress' rather than a primary research article presenting a novel method or discovery. It synthesizes existing advancements in the field rather than introducing groundbreaking innovation.

Commercial Potential

8/10

Structure prediction for adaptive immune receptors is a platform-enabling capability that directly accelerates the design of high-affinity therapeutic antibodies and TCR-T cell therapies, representing a massive commercial market in precision medicine.

Market Size

9/10

The ability to predict and design adaptive immune receptors is a platform-enabling technology with massive addressable demand across biologics, personalized cancer vaccines, and autoimmune therapies.

Tech Readiness

3/10

The title indicates a review or research progress report on structure prediction and design, which is fundamentally a computational/foundational stage of development. It focuses on modeling and design principles rather than a validated clinical candidate or a commercialized diagnostic platform.

IP Strength

6/10

The work focuses on structure prediction and design for adaptive immune receptors, which provides a strong foundation for platform-based IP in biologics; however, without a specific novel algorithm or proprietary dataset mentioned, it likely represents incremental progress in an open-science field.

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7. Phosphoproteomic profiling reveals reversal of dysregulated kinase signaling **28/50** by nitric oxide inhibition in the Shank3 mouse model of autism.

"Discovery identifies specific molecular targets reversing autism-like symptoms, paving the way for precision therapies."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

Institute for Drug Research, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Haitham.amal@mail.huji.ac.il; The Rosamund Stone Zander and Hansjoerg Wyss Translational Neuroscience Center, Boston Children's Hospital, Boston, MA, USA. Haitham.amal@mail.huji.ac.il.

Published in Scientific reports · 2026-06

ABOUT THIS RESEARCH

Researchers used phosphoproteomic profiling to map how nitric oxide inhibition restores normal signaling pathways in a mouse model of autism related to the SHANK3 gene. The study identifies specific protein phosphorylation changes that could explain how targeting nitric oxide synthase can reverse behavioral and synaptic symptoms.

COMMERCIAL ANGLE

The findings provide high-resolution molecular targets for the development of precision pharmacological interventions aimed at correcting dysregulated kinase signaling in autism spectrum disorders.

COMMERCIALISATION METRICS

Novelty

6/10

The study provides meaningful mechanistic insight by mapping the signaling pathways downstream of NO inhibition, but it follows a predictable biological logic rather than introducing a groundbreaking new paradigm or platform technology.

Commercial Potential

6/10

The study identifies a mechanistic target (nNOS inhibition) for a high-unmet-need indication, but the phosphoproteomic focus is primarily descriptive rather than proposing a novel, druggable modality or proprietary platform.

Market Size

9/10

Autism spectrum disorder represents a massive, global therapeutic market with significant unmet medical need and a high prevalence across diverse demographics. Targeting a fundamental neurobiological pathway like nNOS/phosphoproteomics offers potential for high-value, broad-spectrum clinical application.

Tech Readiness

3/10

The research is currently at the fundamental discovery stage, utilizing animal models (Shank3 mice) to identify mechanistic signaling pathways rather than testing a clinical candidate in humans. While it provides high-value biological validation, it lacks the preclinical development or pharmacological optimization required for higher TRL scores.

IP Strength

4/10

The paper describes fundamental mechanism-of-action (MoA) discovery rather than a novel composition of matter; while identifying signaling pathways is valuable for target validation, the use of nitric oxide inhibition is an established biological phenomenon that offers limited new patentable space.

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8. Plaque Reduction Efficacy of an Advanced Electric Toothbrush Compared with a Manual Toothbrush Among 6- to 10-Year-Old Children: Results From a 4-Week Randomised Controlled Trial. 28/50

"Clinical trial validates advanced electric toothbrush efficacy for children, bolstering pediatric oral care market expansion."

Main researcher: Esti Davidovich

Faculty of Dental Medicine, Hebrew University & Hadassah, Jerusalem, Israel.

Published in International journal of paediatric dentistry · 2026 Jul 1

ABOUT THIS RESEARCH

This clinical study demonstrates that an advanced oscillating-rotating electric toothbrush is significantly more effective at reducing plaque in children aged 6-10 than a manual toothbrush. The trial showed a 51% greater whole-mouth plaque reduction over a four-week period.

COMMERCIAL ANGLE

The findings support the commercial expansion of specialized pediatric oral care devices and high-end consumer dental hygiene technology targeted at the growing pediatric wellness market.

COMMERCIALISATION METRICS

Novelty

2/10

The study evaluates an existing commercial consumer product (Oral-B iO2) against a standard manual toothbrush, offering incremental clinical validation rather than scientific or technological innovation. There is no new mechanism, platform technology, or groundbreaking discovery presented that departs from established dental hygiene principles.

Commercial Potential

9/10

The study provides clinical validation for an existing, high-growth product category (pediatric oral care) using a specific, market-ready device (Oral-B iO2). The statistically significant efficacy data directly supports premium pricing and market penetration strategies for a mass-market consumer health product.

Market Size

6/10

The target market is a specific niche (pediatric oral hygiene) within the broader consumer dental care industry, offering steady demand but limited premium expansion compared to platform technologies. While the clinical efficacy supports market entry, the product is a specialized consumer good rather than a high-margin medical device or a broad-scale platform technology.

Tech Readiness

9/10

The technology is already a commercial product (Oral-B iO2) undergoing late-stage clinical validation to support specific pediatric indications. The study demonstrates high TRL by confirming efficacy in a randomized controlled trial (RCT) setting, moving beyond lab-stage proof-of-concept toward market expansion.

IP Strength

2/10

The paper evaluates a commercially existing product (Oral-B iO2) rather than describing a novel, patentable invention or underlying technological breakthrough. The study demonstrates clinical efficacy for a known device, which provides market validation but offers no evidence of new, defensible intellectual property.

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